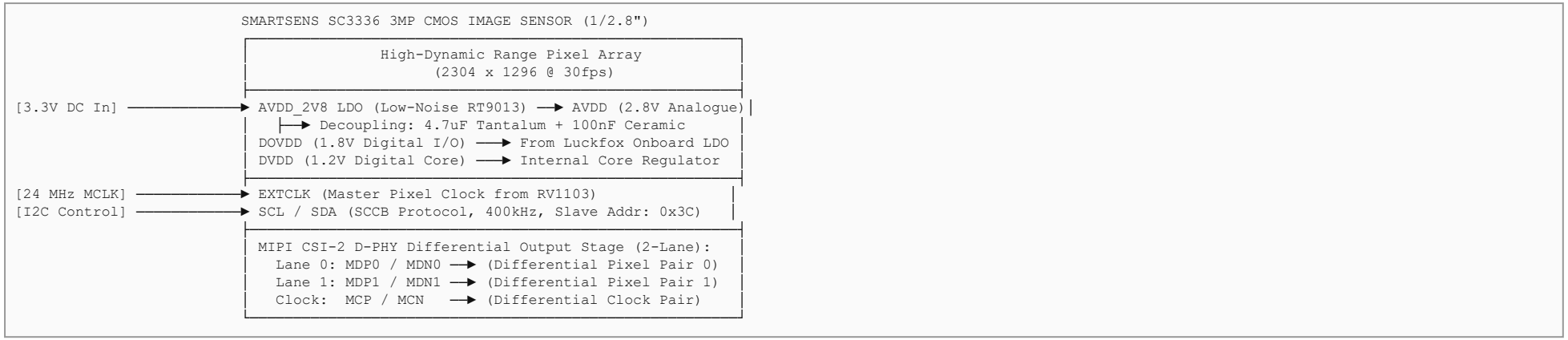
CAMERA MODULE SCHEMATICS: SMARTSENS SC3336 (SHEET 1)

1. OPTICAL SENSOR CORE CIRCUIT & ARCHITECTURE



2. OPTICAL & ELECTRICAL OPERATING PARAMETERS

PARAMETER	SPECIFICATION	ENGINEERING NOTE
Optical Format	1/2.8-inch Progressive Scan CMOS	High-sensitivity pixel architecture for automotive night vision.
Active Resolution	2304 (H) x 1296 (V) pixels (3.0 Megapixels)	Captures 1080p Full HD with zero scaling loss and 2K capability.
Pixel Size	2.0 um x 2.0 um BSI Technology	High signal-to-noise ratio (SNR) with low dark-current noise.
Output Interface	MIPI CSI-2 (2-lane, D-PHY specification)	Differential signaling up to 800 Mbps per data lane.
Supply Rails	AVDD: 2.8V, DOVDD: 1.8V, DVDD: 1.2V	Filtered via low-noise LDOs and ferrite bead decoupling.

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-CAM-SCH-001	DATE 2026-10-08	SHEET 1 OF 2
DOCUMENT TITLE CAMERA SCHEMATICS SPECIFICATION	REV 1.4	DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824	
		ENGINEERING DISCIPLINE OPTICAL SENSORS & SCHEMATICS	STATUS RELEASED	

CAMERA MODULE SCHEMATICS: FPC PINOUT & INTERCONNECT (SHEET 2)

3. 15-PIN 0.5MM PITCH FPC CONNECTOR PIN ASSIGNMENT & SIGNAL DEFINITION

PIN	PIN NAME	SIGNAL TYPE	VOLTAGE LEVEL	DESCRIPTION & INTERCONNECT DETAILS
1	GND	Power Ground	0V	System Ground return and shielding plane.
2	MIPI_D0_N	MIPI D-PHY Out	+/- 200mV Diff	MIPI CSI-2 Data Lane 0 Negative Differential Signal.
3	MIPI_D0_P	MIPI D-PHY Out	+/- 200mV Diff	MIPI CSI-2 Data Lane 0 Positive Differential Signal.
4	GND	Power Ground	0V	Isolation ground between Data 0 and Clock lanes.
5	MIPI_CLK_N	MIPI D-PHY Out	+/- 200mV Diff	MIPI CSI-2 Clock Negative Differential Signal.
6	MIPI_CLK_P	MIPI D-PHY Out	+/- 200mV Diff	MIPI CSI-2 Clock Positive Differential Signal.
7	GND	Power Ground	0V	Isolation ground between Clock and Data 1 lanes.
8	MIPI_D1_N	MIPI D-PHY Out	+/- 200mV Diff	MIPI CSI-2 Data Lane 1 Negative Differential Signal.
9	MIPI_D1_P	MIPI D-PHY Out	+/- 200mV Diff	MIPI CSI-2 Data Lane 1 Positive Differential Signal.
10	GND	Power Ground	0V	Digital signal ground shielding plane.
11	CAM_RSTn	Digital Input	3.3V / 1.8V	Active-low sensor hardware reset line (driven by RV1103).
12	CAM_PDN	Digital Input	3.3V / 1.8V	Sensor power-down control line (pulled low for active state).
13	I2C_SCL	Bidirectional	3.3V Open-Drain	SCCB register programming clock with 2.2k pull-up to 3.3V.
14	I2C_SDA	Bidirectional	3.3V Open-Drain	SCCB register programming data with 2.2k pull-up to 3.3V.
15	VCC_3V3	Power Feed	3.3V DC (+/- 5%)	Main power feed to onboard 2.8V AVDD low-noise LDO regulator.

4. SIGNAL INTEGRITY & LAYOUT CONSTRAINTS

- Differential impedance for MIPI data and clock pairs: 100 Ohm (+/- 10%) differential.
- Intra-pair length matching: skew less than 0.5mm between positive and negative traces.
- Dedicated solid ground reference plane directly beneath all MIPI differential routing.

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-CAM-SCH-001	DATE 2026-10-08	SHEET 2 OF 2	
DOCUMENT TITLE CAMERA SCHEMATICS SPECIFICATION	REV 1.4	DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824		
		ENGINEERING DISCIPLINE OPTICAL SENSORS & SCHEMATICS	STATUS RELEASED		